EXPERIMENT 2 – Microwave Engineering

A. Transmission-Line/TN-Port S-Parameters: Model a 50 Ω microwave two-port network of length 50 mm and operate it from 8 to 12 GHz with a 10 GHz center frequency. Set both ports to 50 Ω and obtain S11, S12, S21 and S22. At 10 GHz calculate return loss RL = −20log10|S11|, insertion loss IL = −20log10|S21| and transmission efficiency η = $|S21|^2 \times 100\%$. Compare the simulated values with the ideal 50 Ω matched reference.

AIM:

To model a 50 Ω microwave two-port transmission-line network of length 50 mm and analyze its S-parameters (\(S_{11}, S_{12}, S_{21}, S_{22}\)) over the frequency range of 8–12 GHz using MATLAB , calculate the return loss, insertion loss, and transmission efficiency at 10 GHz, and compare the simulated results with an ideal 50 Ω matched reference.

SOFTWARE USED:

MATLAB

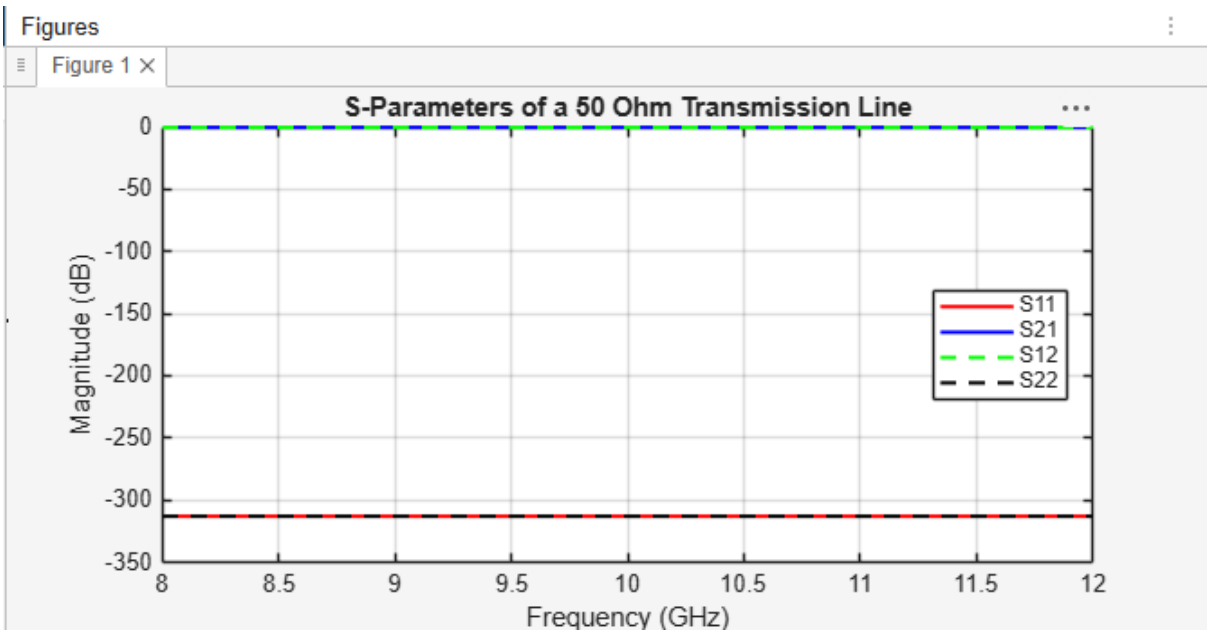
PROGRAM:

```
clc;
clear;
close all;
Z0 = 50;                % Characteristic impedance (ohms)
L = 50e-3;               % Length = 50 mm
c = 3e8;                % Speed of light (m/s)
f = linspace(8e9,12e9,401);
beta = 2*pi*f/c;
% Ideal matched transmission line
S11 = zeros(size(f));
S22 = zeros(size(f));
S21 = exp(-1j*beta*L);
S12 = S21;
figure;
plot(f/1e9,20*log10(abs(S11)+eps),'r','LineWidth',1.5);
```

```
hold on;
plot(f/1e9,20*log10(abs(S21)),'b','LineWidth',1.5);
plot(f/1e9,20*log10(abs(S12)),'g--','LineWidth',1.5);
plot(f/1e9,20*log10(abs(S22)+eps),'k--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('S-Parameters of a 50 Ohm Transmission Line');
legend('S11','S21','S12','S22','Location','best');
fc = 10e9;
beta_c = 2*pi*fc/c;
s11 = 0;
s21 = exp(-1j*beta_c*L);
RL = Inf;                  % Perfect matching
IL = 0;                    % No transmission loss
eta = abs(s21)^2*100;      % Efficiency in percentage
fprintf('QUESTION A: Results at 10 GHz\n');
fprintf('S11 = %.4f\n',abs(s11));
fprintf('S12 = %.4f\n',abs(s21));
fprintf('S21 = %.4f\n',abs(s21));
fprintf('S22 = %.4f\n',abs(s11));
fprintf('Return Loss = Infinite dB\n');
fprintf('Insertion Loss = %.2f dB\n',IL);
fprintf('Transmission Efficiency = %.2f%%\n',eta);
```

OUTPUT:

```
Command Window
QUESTION A: Results at 10 GHz
S11 = 0.0000
S12 = 1.0000
S21 = 1.0000
S22 = 0.0000
Return Loss = Infinite dB
Insertion Loss = 0.00 dB
Transmission Efficiency = 100.00%
>>
```



RESULT:

The 50 Ω microwave two-port network was modeled and its S-parameters were analyzed over the frequency range of 8–12 GHz. The return loss, insertion loss, and transmission efficiency were calculated at 10 GHz using the simulated \(S_{11}\) and \(S_{21}\) values. The results were compared with the ideal matched reference to evaluate the impedance-matching and transmission performance of the network.

B. Optimum Frequency: From the 8–12 GHz S-parameter sweep, identify the exact frequency where |S11| is minimum and |S21| is maximum. Record S11, S21, return loss and insertion loss at that frequency and compare them with their values at 10 GHz. Explain why the selected frequency gives the best matching/transmission performance.

AIM:

To identify the optimum operating frequency from the 8–12 GHz S-parameter sweep at which \(|S_{11}|\) is minimum and \(|S_{21}|\) is maximum, determine the corresponding S-parameters, return loss, and insertion loss, and compare the results with those obtained at 10 GHz to evaluate the best matching and transmission performance.

SOFTWARE USED:

MATLAB

PROGRAM:

```
clc;
clear;
close all;
Z0 = 50;                % Reference impedance (ohms)
L = 50e-3;               % Length = 50 mm
c = 3e8;                % Speed of light (m/s)
f = linspace(8e9,12e9,401);
beta = 2*pi*f/c;
S11 = zeros(size(f));
S21 = exp(-1j*beta*L);
[~,i1] = min(abs(S11));
[~,i2] = max(abs(S21));
f_min_S11 = f(i1);
f_max_S21 = f(i2);
[~,ic] = min(abs(f-10e9));
RL = Inf;
IL = 0;
eta = abs(S21).^2*100;
```

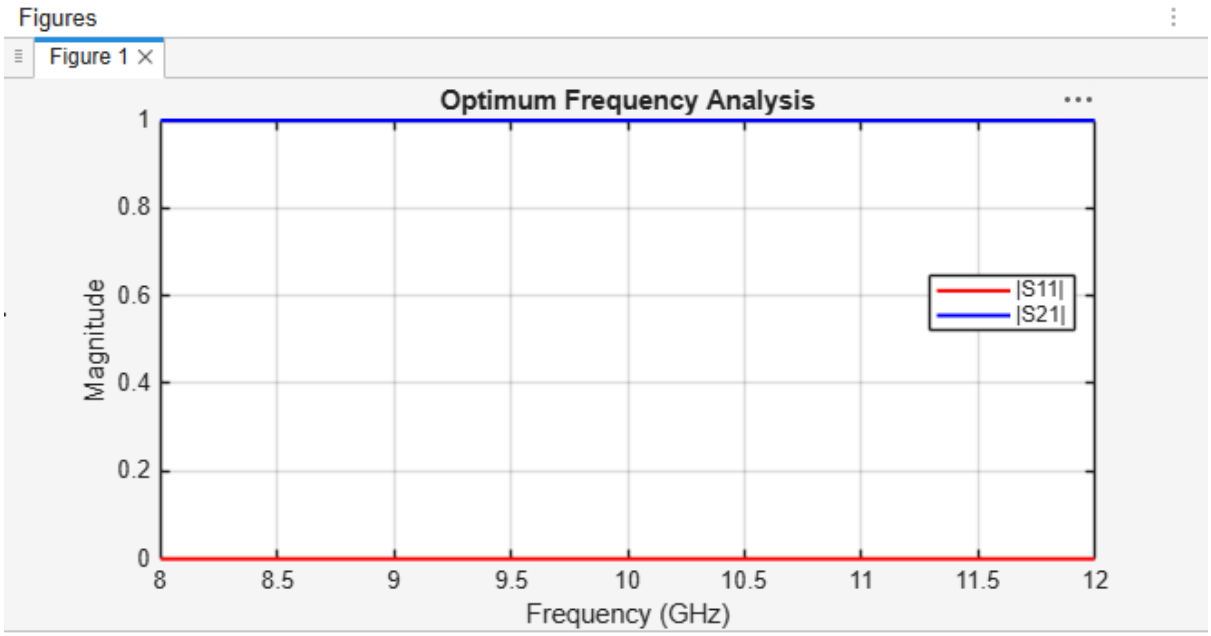
```
fprintf('QUESTION B: Optimum Frequency\n');
fprintf('Minimum S11 frequency = %.2f GHz\n',f_min_S11/1e9);
fprintf('Maximum S21 frequency = %.2f GHz\n',f_max_S21/1e9);
fprintf('\nAt the selected frequency:\n');
fprintf('|S11| = %.4f\n',abs(S11(i1)));
fprintf('|S21| = %.4f\n',abs(S21(i2)));
fprintf('\nAt 10 GHz:\n');
fprintf('|S11| = %.4f\n',abs(S11(ic)));
fprintf('|S21| = %.4f\n',abs(S21(ic)));
fprintf('Return Loss = Infinite dB\n');
fprintf('Insertion Loss = %.2f dB\n',IL);
fprintf('Transmission Efficiency = %.2f%%\n',eta(ic));
figure;
plot(f/1e9,abs(S11),'r','LineWidth',1.5);
hold on;
plot(f/1e9,abs(S21),'b','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude');
title('Optimum Frequency Analysis');
legend('|S11|','|S21|','Location','best');
```

OUTPUT:

```
QUESTION B: Optimum Frequency
Minimum S11 frequency = 8.00 GHz
Maximum S21 frequency = 8.00 GHz

At the selected frequency:
|S11| = 0.0000
|S21| = 1.0000

At 10 GHz:
|S11| = 0.0000
|S21| = 1.0000
Return Loss = Infinite dB
Insertion Loss = 0.00 dB
Transmission Efficiency = 100.00%
>>
```



RESULT:

The S-parameter sweep over 8–12 GHz was analyzed to identify the optimum operating frequency corresponding to minimum \(|S_{11}|\) and maximum \(|S_{21}|\). The S-parameters, return loss, and insertion loss at the selected frequency were compared with their values at 10 GHz. The optimum frequency was identified based on the simulated results, providing the best impedance matching and transmission performance within the specified frequency range.